

tf.compat.v1.distributions.Dirichlet Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/distributions/Dirichlet

Dirichlet distribution.

Function Signatures

```
tf.compat.v1.distributions.Dirichlet( concentration,  
    validate_args=False, allow_nan_stats=True, name='Dirichlet' )  
(k-1)
```

Code Examples

```
tf.compat.v1.distributions.Dirichlet(  
    concentration,  
    validate_args=False,  
    allow_nan_stats=True,  
    name='Dirichlet'  
)
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    concentration,  
    validate_args=False,  
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)
```

$S^{k-1} = \{ (x_0, \dots, x_{k-1}) \text{ in } \mathbb{R}^k : \sum_j x_j = 1 \text{ and all}_j x_j > 0 \}.$

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```
pdf(x; alpha) = prod_j x_j**(alpha_j - 1) / Z  
Z = prod_j Gamma(alpha_j) / Gamma(sum_j alpha_j)
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concentration = alpha = mean * total_concentration
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Documentation

Dirichlet distribution.

Inherits From: Distribution

The Dirichlet distribution is defined over the $(k-1)$ -simplex using a positive, length- k vector concentration ($k > 1$). The Dirichlet is identically the Beta distribution when $k = 2$.

Mathematical Details

The Dirichlet is a distribution over the open $(k-1)$ -simplex, i.e.,

The probability density function (pdf) is,

where:

- x in S^{k-1} , i.e., the $(k-1)$ -simplex,
- $\text{concentration} = \alpha = [\alpha_0, \dots, \alpha_{k-1}]$, $\alpha_j > 0$,
- Z is the normalization constant aka the multivariate beta function,

and,

- Γ is the gamma function.

The concentration represents mean total counts of class occurrence, i.e.,

where mean in $S^{\{k-1\}}$ and total_concentration is a positive real number representing a mean total count.

Distribution parameters are automatically broadcast in all functions; see examples for details.

Samples of this distribution are reparameterized (pathwise differentiable). The derivatives are computed using the approach described in (Figurnov et al., 2018).

Examples

Compute the gradients of samples w.r.t. the parameters:

References

Attributes

Stats return +/- infinity when it makes sense. E.g., the variance of a Cauchy distribution is infinity. However, sometimes the statistic is undefined, e.g., if a distribution's pdf does not achieve a maximum within the support of the distribution, the mode is undefined. If the mean is undefined, then by definition the variance is undefined. E.g. the mean for Student's T for $df = 1$ is undefined (no clear way to say it is either + or - infinity), so the variance = $E[(X - \text{mean})^2]$ is also undefined.

May be partially defined or unknown.

The batch dimensions are indexes into independent, non-identical parameterizations of this distribution.

May be partially defined or unknown.

Currently this is one of the static instances
distributions.FULLY_REPARAMETERIZED
or distributions.NOT_REPARAMETERIZED.

Methods

`## batch_shape_tensor`

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Shape of a single sample from a single event index as a 1-D Tensor.

The batch dimensions are indexes into independent, non-identical
parameterizations of this distribution.

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Cumulative distribution function.

Given random variable X , the cumulative distribution function cdf is:

[View source](#)

Creates a deep copy of the distribution.

covariance

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Covariance.

Covariance is (possibly) defined only for non-scalar-event distributions.

For example, for a length- k , vector-valued distribution, it is calculated as,

where Cov is a (batch of) $k \times k$ matrix, $0 \leq (i, j) < k$, and E denotes expectation.

Alternatively, for non-vector, multivariate distributions (e.g., matrix-valued, Wishart), Covariance shall return a (batch of) matrices under some vectorization of the events, i.e.,

where Cov is a (batch of) $k' \times k'$ matrices, $0 \leq (i, j) < k' = \text{reduce_prod}(\text{event_shape})$, and Vec is some function mapping indices of this distribution's event dimensions to indices of a length- k' vector.

cross_entropy

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Computes the (Shannon) cross entropy.

Denote this distribution (self) by P and the other distribution by Q . Assuming P, Q are absolutely continuous with respect to one another and permit densities $p(x) dx$ and $q(x) dx$, (Shanon) cross entropy is defined as:

where F denotes the support of the random variable $X \sim P$.

entropy

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Shannon entropy in nats.

event_shape_tensor

[View source](#)

Shape of a single sample from a single batch as a 1-D int32 Tensor.

is_scalar_batch

[View source](#)

Indicates that batch_shape == [].

is_scalar_event

[View source](#)

Indicates that event_shape == [].

kl_divergence

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Computes the Kullback--Leibler divergence.

Denote this distribution (self) by p and the other distribution by q . Assuming p, q are absolutely continuous with respect to reference measure r , the KL divergence is defined as:

where F denotes the support of the random variable $X \sim p$, $H[., .]$ denotes (Shanon) cross entropy, and $H[.]$ denotes (Shanon) entropy.

log_cdf

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Log cumulative distribution function.

Given random variable X , the cumulative distribution function cdf is:

Often, a numerical approximation can be used for $\log_cdf(x)$ that yields a more accurate answer than simply taking the logarithm of the cdf when $x \ll -1$.

log_prob

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Log probability density/mass function.

Additional documentation from Dirichlet:

log_survival_function

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Log survival function.

Given random variable X , the survival function is defined:

Typically, different numerical approximations can be used for the log survival function, which are more accurate than $1 - cdf(x)$ when $x \gg 1$.

[View source](#)

[View source](#)

Additional documentation from Dirichlet:

param_shapes

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Shapes of parameters given the desired shape of a call to `sample()`.

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is returned for that instance's call to `sample()`.

Subclasses should override class method `_param_shapes`.

param_static_shapes

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`param_shapes` with static (i.e. `TensorShape`) shapes.

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is returned for that instance's call to `sample()`. Assumes that the sample's shape is known statically.

Subclasses should override class method `_param_shapes` to return constant-valued tensors when constant values are fed.

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Probability density/mass function.

Additional documentation from Dirichlet:

quantile

[View source](#)

Quantile function. Aka "inverse cdf" or "percent point function".

Given random variable X and p in $[0, 1]$, the quantile is:

sample

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Generate samples of the specified shape.

Note that a call to `sample()` without arguments will generate a single sample.

stddev

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